

Report on Plans and Challenges faced

This project was a journey of shifting from standard processing to high-performance engineering, making critical choices about data integrity, and ultimately mastering ensemble techniques to achieve peak predictive power.

1. Data Engineering & The "Merging" Hurdle

My initial challenge was architectural: How to merge massive datasets without crashing the environment?

I started by weighing the pros and cons of using SQL vs. standard Python (Pandas). I eventually discovered Polars, which became a turning point. Its lazy evaluation and multi-threaded execution allowed me to handle the joins significantly faster than traditional methods.

Selective Joining: I had to be extremely disciplined with column selection. Instead of a "blind merge," I manually identified which columns from the auxiliary datasets were essential for the join and which were redundant, ensuring the memory footprint remained manageable.

2. Data Cleaning and EDA Strategy

With a dataset this large, Exploratory Data Analysis (EDA) could easily become overwhelming.

Handling Nulls and NaNs: I moved beyond simple imputation. I spent significant time analyzing the nature of missing data, looking for patterns that might suggest the "absence" of data was a signal in itself.

Statistical Deep-Dive: To select features, I employed numerical correlation matrices and Cramer's V for categorical/ordinal data. This allowed me to identify which features actually held predictive weight and which were just noise.

Visual Selectivity: Because of the sheer volume of data, I had to be selective about which visualizations to generate, focusing on features that showed the highest variance or correlation with the target.

3. The Philosophy of "Data Originality"

One of my most significant strategic pivots was in Feature Engineering:

Preserving Outliers: While common practice often dictates removing outliers, I chose to keep them. I believe in credit risk, the "extremes" often hold the most important signals for default.

Minimalist Feature Creation: I intentionally avoided creating an excessive amount of synthetic columns. My goal was to keep the dataset as "original" as possible, allowing the models to find the latent patterns within the raw data rather than forcing a human-centric interpretation on them.

The "No SMOTE" Decision: Despite the class imbalance, I decided against using SMOTE (Synthetic Minority Over-sampling Technique). I wanted the models to learn from the real distribution of the bank's history(dataset as "original" as possible), opting instead to handle imbalance through Threshold Analysis and model-specific weighting.

4. Modeling and the Ensemble Evolution

The final phase was a steep learning curve in algorithmic diversity:

The Three Pillars: I implemented and tuned three distinct models: Random Forest, XGBoost, and LightGBM. Each brought a different perspective—RF for stability, XGBoost for precision, and LightGBM for efficiency.

Learning the Ensemble: The culmination of the project was learning and implementing an Ensemble strategy. By creating a 50/50 weighted blend of LightGBM and XGBoost, I was able to smooth out individual model errors.

5. Final Results

The success of these choices is reflected in the final metrics:

The Winning Model: The Ensemble (LightGBM + XGBoost) emerged as the top performer, validating my initial architectural strategy. It achieved a robust and highly competitive Final AUC of 0.7850, proving that blending these two gradient-boosting frameworks was the key to maximizing predictive power.

Strategic Outcome: My decision to perform an extensive Threshold Analysis provided a comprehensive view of the model's behavior across the entire probability spectrum. This made it possible to identify 0.48 as the optimal threshold. At this level, the model achieves a 71% Recall for defaults, striking a sophisticated balance: it is aggressive enough to protect the bank's capital from high-risk loans while remaining commercially viable by minimizing unnecessary rejections.

Final Reflection

This project has been a significant milestone in my development as a data scientist. My biggest takeaway is that the 'best' model isn't just a result of the code; it is the outcome of a series of principled 'No's.' By choosing not to use SMOTE, not to remove outliers, and not to clutter the dataset with redundant features, I was able to preserve the purity of the data signal.

The transition from traditional SQL-based merging to the high-performance capabilities of Polars, and ultimately to a fine-tuned Ensemble, transformed this project into a masterclass in end-to-end data science. I've learned that in credit risk, respecting the original distribution of the data is often more powerful than forcing synthetic corrections.

Thank you.